

Movement Dynamics in Elite Female Soccer Athletes: The Quantile Cube Approach

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The Problem: Aggregates Mask Performance



- Wearable devices capture detailed velocity, acceleration, & position information. [1]
- **Problem:** Traditional metrics = aggregates or arbitrary thresholds



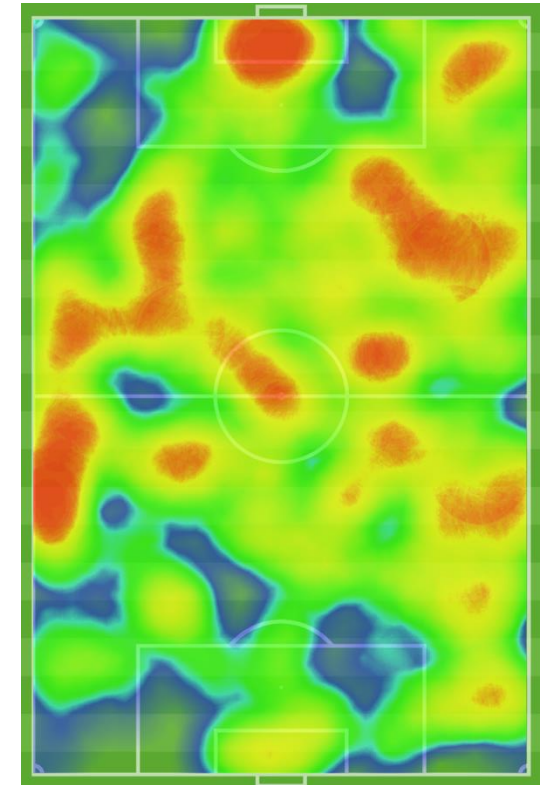
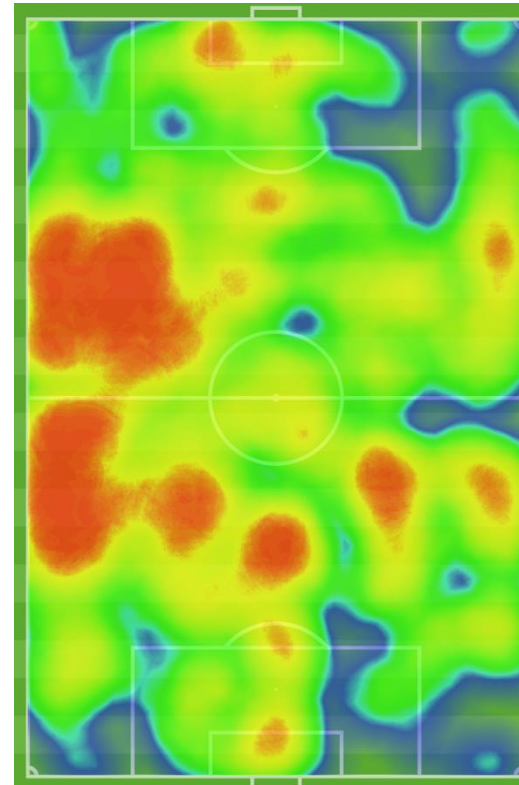
[1]

- Movement patterns and workloads vary across positions [2, 3], match periods [4, 5], genders [6, 3], competitive contexts [7], playing levels [8, 9, 10], environmental conditions [11]

The Solution: Moving Beyond the Average



- Effective athlete monitoring requires interpretable, data-driven models.
- **Goal:** Build a framework that captures the *distribution* of movement
- **Solution:** The Quantile Cube



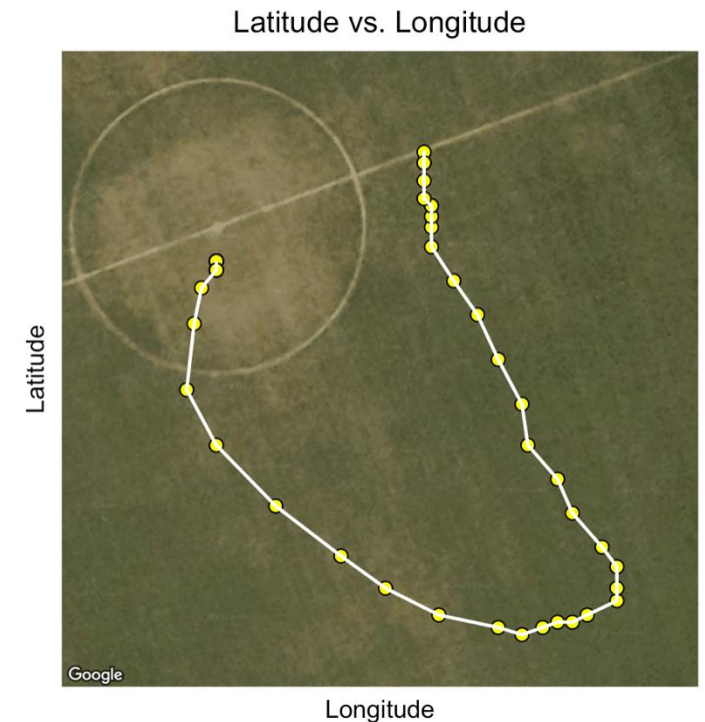
The Data & Processing



- Raw GPS Data filtered to 9 female athletes across 23 matches
 - 1Hz sample of timestamp, latitude, and longitude
 - **Total:** 198 athlete-matches → 396 athlete-match-halves

For each athlete-match-half, we

1. Converted longitude and latitude to meters ^[12]
2. Fitted 3rd-degree interpolating spline at 10Hz
3. Derived velocity (m/s), acceleration (m/s²), and angle

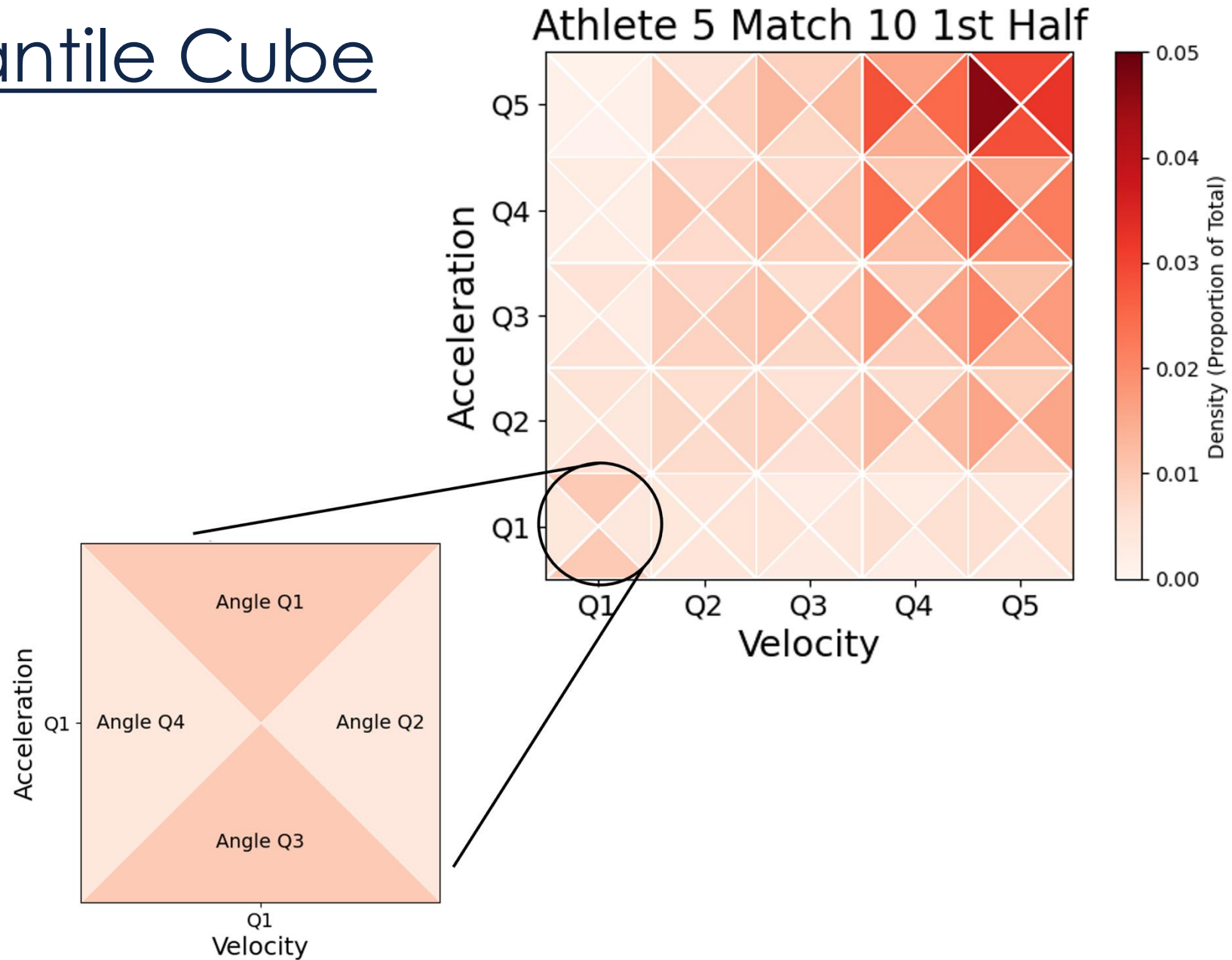


The Quantile Cube



- A 3-D summary representation of athlete movement distributions
- **Construction**
 1. **Derived quantile cutoffs using the full dataset**
 - 5 quantiles for Velocity and Acceleration
 - 4 quantiles for angle, starting from a shifted -30°
 2. **Constructed a quantile cube for each athlete-match-half**
 - 100-dimensional vector representing either deciseconds of time or proportion of total time spent in a certain movement category
 3. **Result: $Y_{n \times d}$**
 - $n = 396$ rows: quantile cube for an athlete-match-half
 - $d = 100$ features: time spent within the corresponding movement profile

The Quantile Cube



Covariates



- Covariates for each athlete-match-half: $\mathbf{X}_{n \times r}$
 - $n = 396$ rows: athlete-match-half
 - $r = 13$ covariates:
 - **Match-level (10):**
 - Match ID
 - location (home, away, neutral)
 - half (1st or 2nd)
 - result (win, loss, tie)
 - goals scored and conceded (halftime and full time)
 - score differentials (halftime and full time)
 - **Athlete-level (3):**
 - Athlete ID
 - position (defender, midfield, forward)
 - playing time (by half)

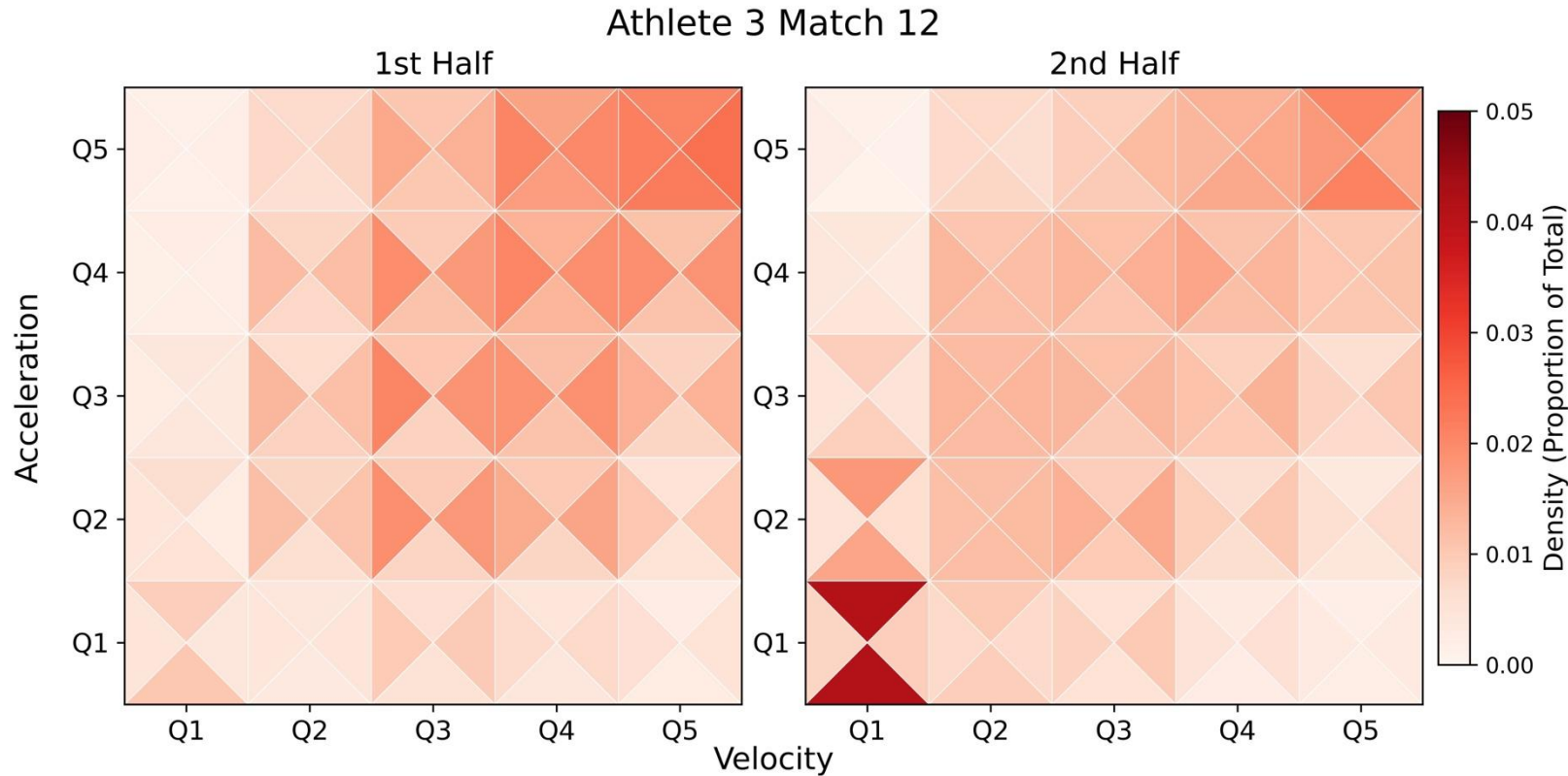
Methods and Results Overview



Three-step Process:

1. **Quantified Differences** in movement distributions between match halves
2. **Reduced dimensionality** of movement patterns in match contexts
3. **Modeled movement distributions** as functions of the covariates

Quantifying Differences



- **Issue:** high dimensionality and complex dependence structures
- **Solution:** Hellinger distance metric [13]

Resampling-Based
Hypothesis Testing

+

Multiple Testing
Correction



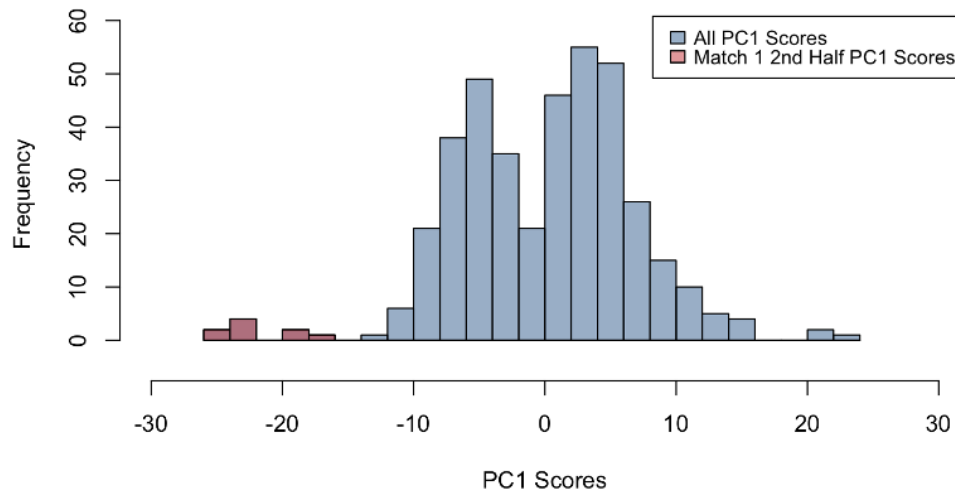
**Significant distributional
differences for *almost*
every athlete-match pair.**

Reducing Dimensionality



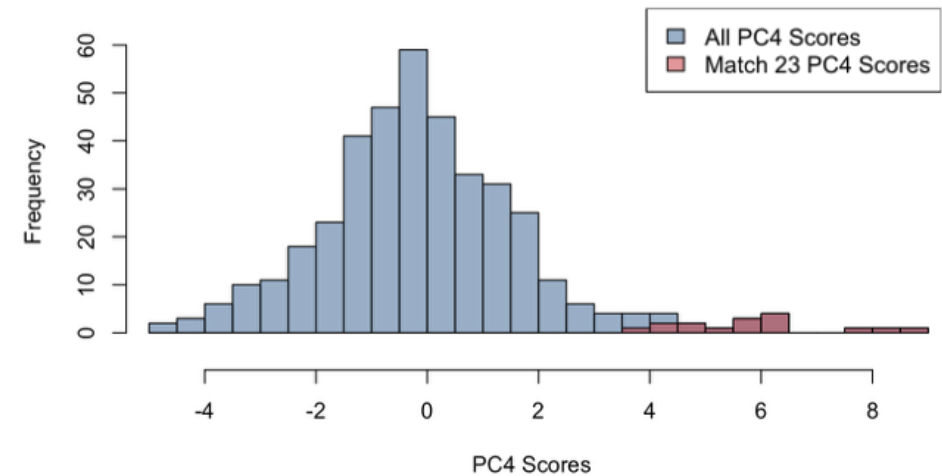
- Applied Principal Component Analysis (PCA) to $\mathbf{Y}_{n \times d}$
 - **Goal:** Capture key variation to explain movement patterns across matches and athletes
- Variance Explained $\neq$ Useful Interpretation
 - Component scores were plotted against match- and athlete-level characteristics.

Histogram of PC1 Scores



PC1: less time in the middle quantiles of velocity and acceleration during the 2nd half of Match 1

Histogram of PC4 Scores

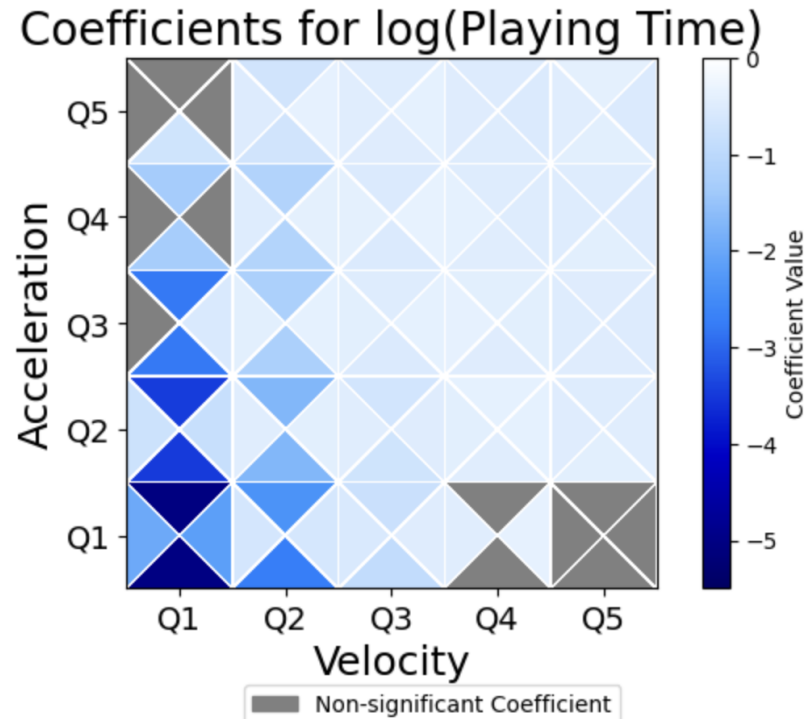
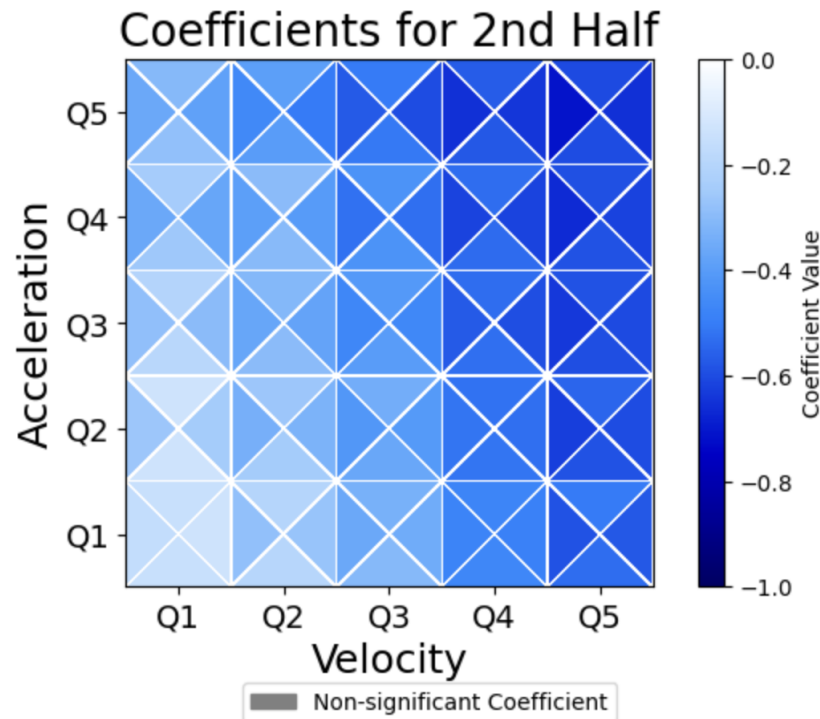


PC4: increased time spent in low-velocity, high-acceleration movements in Match 23

Modeling Movement Distributions



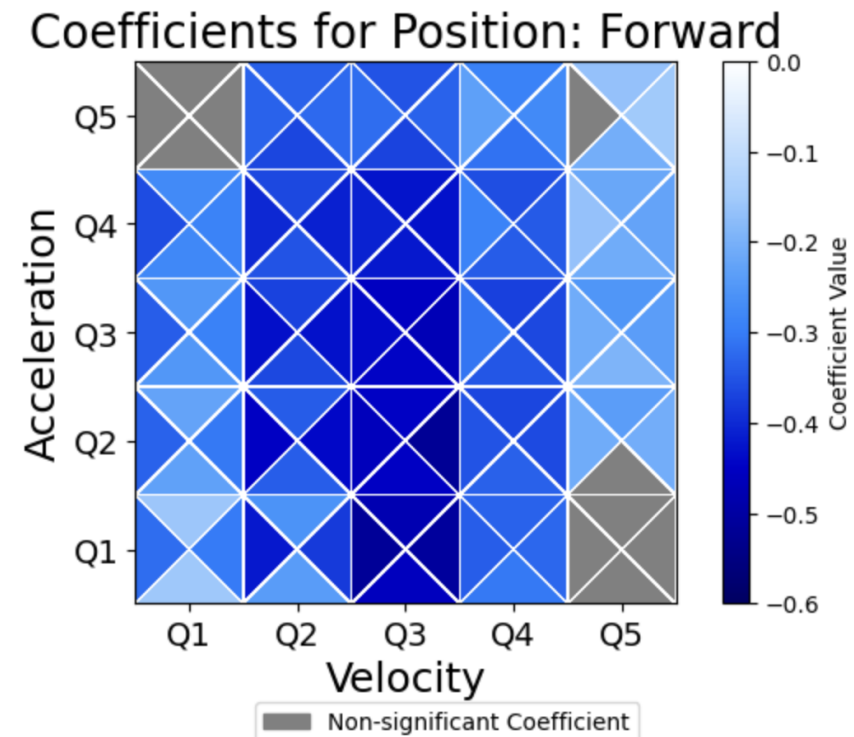
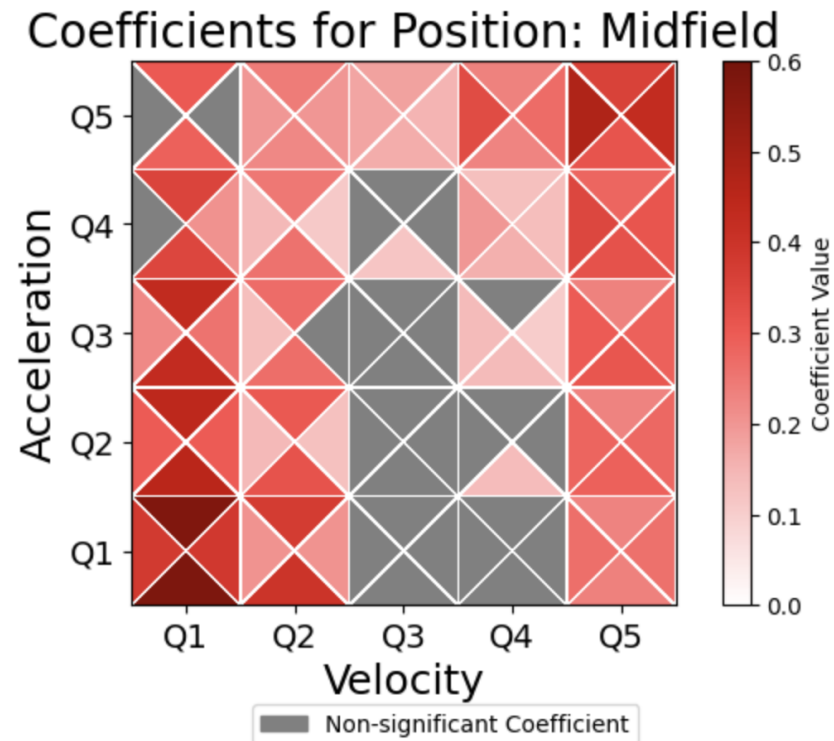
- **Goal:** Relate athlete movement distributions to contextual factors using Dirichlet-multinomial regression (DMR) [14, 15]
- Optimal model: match half, player position, mean-centered $\log(\text{playing time})$



Modeling Movement Distributions



- Optimal model: match half, player position, mean-centered $\log(\text{playing time})$
- Positional differences exhibited clear and interpretable patterns compared to defenders (reference category):



Analytical Extensions



- Fuse multimodal data and synchronize datasets
- Develop real-time, actionable metrics and alert systems for athlete monitoring.
- Collaborate with coaching and performance staff for practical feedback

Filters

Cube 1

Select Position(s):

Position 1

Select Day(s):

Date 12

Select Athlete(s):

Select All x

Select Period(s):

Half 1 x

Cube 2

Select Position(s):

Position 2

Select Day(s):

Date 12

Select Athlete(s):

Select All x

Select Period(s):

Half 1 x

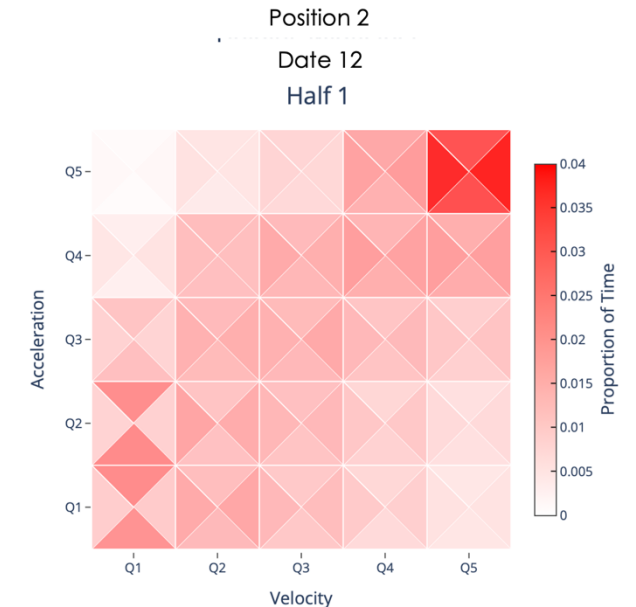
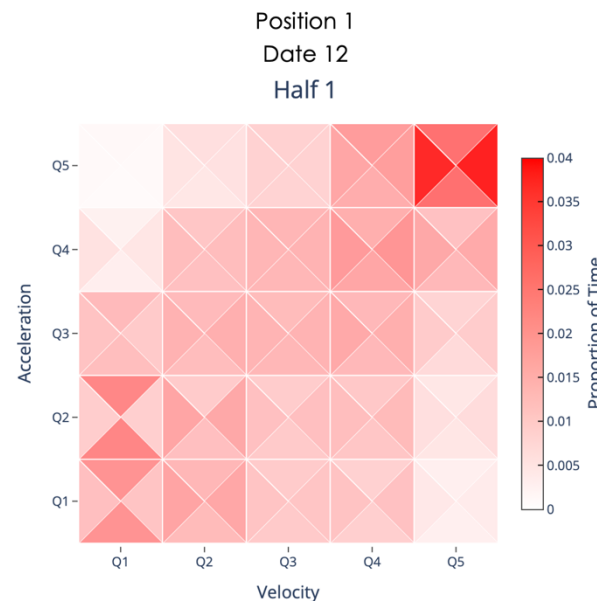
How to Read the Cubes

Each cube shows the breakdown of time spent in different movement types. Darker color means a greater share of time in that movement type. Every value represents the proportion of total time spent in that type — all values together add up to 1. You can multiply each number by 100 to read it as a percentage of total movement time.

Velocity and Acceleration Zones

Each zone (Q1-Q5) represents a range of movement intensity. The value at the top end of a range is *not* included in that zone. For example, 0.22 m/s is part of Q2, not Q1. Inside each Q-by-Q velocity-acceleration cell, the four triangles represent movement direction: forward (top triangle), right (right triangle), backward (bottom triangle), and left (left triangle).

Metric	Q1	Q2	Q3	Q4	Q5
Velocity (m/s)	QUANTILE VALUES				
Acceleration (m/s ²)					



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Thank you! Questions?



Full Paper

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quantile-cube Python Package Now Available on PyPI!

References



1. Cloe Cummins, Rhonda Orr, Helen O'Connor, and Cameron West. Global positioning systems (gps) and microtechnology sensors in team sports: A systematic review. *Sports Medicine*, 43(10):1025–1042, October 2013. doi: 10.1007/s40279-013-0069-2.
2. George M. Wehbe, Timothy B. Hartwig, and Craig S. Duncan. Movement analysis of australian national league soccer players using global positioning system technology. *Journal of Strength and Conditioning Research*, 28(3):834–842, March 2014. doi: 10.1519/JSC.0b013e3182a35dd1.
3. N. M. Kuhlman, A. R. Jagim, M. T. Jones, M. K. Feit, and J. B. Fields. A comparison of match external load demands across women's collegiate field sports. *Journal of Strength and Conditioning Research*, 39(2): 234–241, 2025. doi: 10.1519/JSC.0000000000004960. Epub 2024 Oct 24.
4. Joel Barrera, Hugo Sarmiento, Filipe Manuel Clemente, Adam Field, and António J. Figueiredo. The effect of contextual variables on match performance across different playing positions in professional portuguese soccer players. *International Journal of Environmental Research and Public Health*, 18:5175, May 2021. doi: 10.3390/ijerph18105175.
5. B. J. Snyder, C. Maung-Maung, and C. Whitacre. Indicators of fatigue during a soccer match simulation using gps-derived workload values: Which metrics are most useful? *Sports*, 12(1):9, 2024. doi:10.3390/sports12010009.
6. Bianca J. De Lucia, Mary K. Feit, Ashima Fillbach, and Jennifer B. Fields. Comparison of game external loads across a men's and women's soccer season. *Medicine & Science in Sports & Exercise*, 56(10S):198, 2024. doi: 10.1249/01.mss.0001054244.02429.29.
7. B. N. Bozzini, B. A. McFadden, A. J. Walker, and S. M. Arent. Varying demands and quality of play between in-conference and out-of-conference games in division i collegiate women's soccer. *Journal of Strength and Conditioning Research*, 34(12):3364–3368, December 2020. doi: 10.1519/JSC.00000000000003841.
8. Jocelyn K. Mara, Kevin G. Thompson, Kate L. Pumpa, and Stuart Morgan. Quantifying the high-speed running and sprinting profiles of elite female soccer players during competitive matches using an optical player tracking system. *Journal of Strength and Conditioning Research*, 31(6):1500–1508, June 2017. doi: 10.1519/JSC.0000000000001629.

References



9. Jesse Griffin, Timothy Newans, Sean Horan, Justin Keogh, Melissa Andreatta, and Clare Minahan. Acceleration and high-speed running profiles of women's international and domestic football matches. *Frontiers in Sports and Active Living*, 3, 2021. ISSN 2624-9367. doi: 10.3389/fspor.2021.604605. URL <https://www.frontiersin.org/articles/10.3389/fspor.2021.604605>.
10. N. Datson, B. Drust, M. Weston, I. H. Jarman, P. J. Lisboa, and W. Gregson. Match physical performance of elite female soccer players during international competition. *Journal of Strength and Conditioning Research*, 31(9):2379–2387, September 2017. doi: 10.1519/JSC.0000000000001575.
11. J. D. Bohner, J. R. Hoffman, W. P. McCormack, T. C. Scanlon, J. R. Townsend, J. R. Stout, M. S. Fragala, and D. H. Fukuda. Moderate altitude affects high intensity running performance in a collegiate women's soccer game. *Journal of Human Kinetics*, 47:147–154, October 2015. doi: 10.1515/hukin-2015-0070.
12. Edzer Pebesma. Simple features for r: Standardized support for spatial vector data. *The R Journal*, 10(1): 439–446, 2018. doi: 10.32614/RJ-2018-009.
13. A. W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 1 edition, October 1998. ISBN 9780511802256. doi: 10.1017/CBO9780511802256. URL <https://www.cambridge.org/core/product/identifier/9780511802256/type/book>.
14. James E. Mosimann. On the compound multinomial distribution, the multivariate β distribution, and correlations among proportions. *Biometrika*, 49(1/2):65–82, 1962. ISSN 0006-3444. doi: 10.2307/2333468. Publisher: [Oxford University Press, Biometrika Trust].
15. Jun Chen and Hongzhe Li. Variable selection for sparse dirichlet-multinomial regression with an application to microbiome data analysis. *The Annals of Applied Statistics*, 7(1), March 2013. ISSN 1932-6157. doi: 10.1214/12-AOAS592.
16. J. D. Vescovi and T. G. Favero. Motion characteristics of women's college soccer matches: Female athletes in motion (faim) study. *International Journal of Sports Physiology and Performance*, 9(3):405–414, May 2014. doi: 10.1123/IJSPP.2013-0526.

References



17. Anna E. Saw, Luana C. Main, and Paul B. Gatin. Monitoring the athlete training response: subjective self-reported measures trump commonly used objective measures: a systematic review. *British Journal of Sports Medicine*, 50(5):281–291, 2016. doi: 10.1136/bjsports-2015-094758.
18. V. Mudeng, I. M. Hakim, S. S. Suprpto, and S. W. Choe. An alternative athlete monitoring system using cost-effective inertial sensing instrumentation. *Journal of Electrical Engineering & Technology*, 17(6): 3581–3592, 2022. doi: 10.1007/s42835-022-01258-1. Epub 2022 Sep 26.

Image Sources

1. Catapult GPS. <https://simplifaster.com/articles/catapult-gps/>, accessed September 2025.
2. Gainbridge Super League: Carolina Ascent FC vs. Tampa Bay Sun FC. <https://www.gainbridgesuperleague.com/carolinaascentfc-tampabaysunfc-2554347/> accessed September 2025.